B16 Series B

Tumour scenarios: inferCNV and calling malignant cells

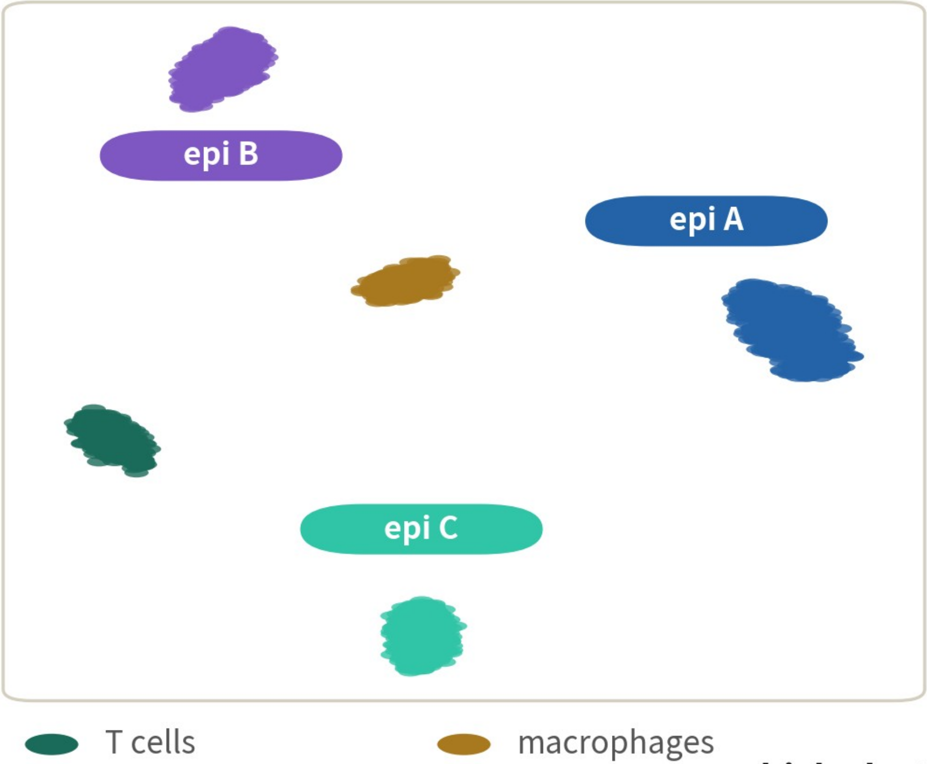
Separate malignant from normal cells with CNV signal — and know when the inference lies to you.

About 38 min | Prerequisites: B5-B10 | Data: infercnv built-in oligodendroglioma example

Three epithelial clusters — which is cancer?

simulated data

clusters: epithelium splits in three



EPCAM: all three are positive

the marker says epithelial — not malignant

which cluster is malignant?

One tumour sample: three epithelial clusters, all EPCAM-positive. Clustering cannot help you

WHERE THIS EPISODE STARTS

Malignancy is defined in the genome — clustering and markers only see the transcriptome.

To call malignancy you must dig the genome's shadow out of the RNA. That is exactly what inferCNV does.

What you will learn

1. Explain why clustering and markers cannot settle malignant vs normal in tumour data

2. Understand inferCNV's chain (genome ordering, sliding windows, reference-relative shifts) and run the built-in example

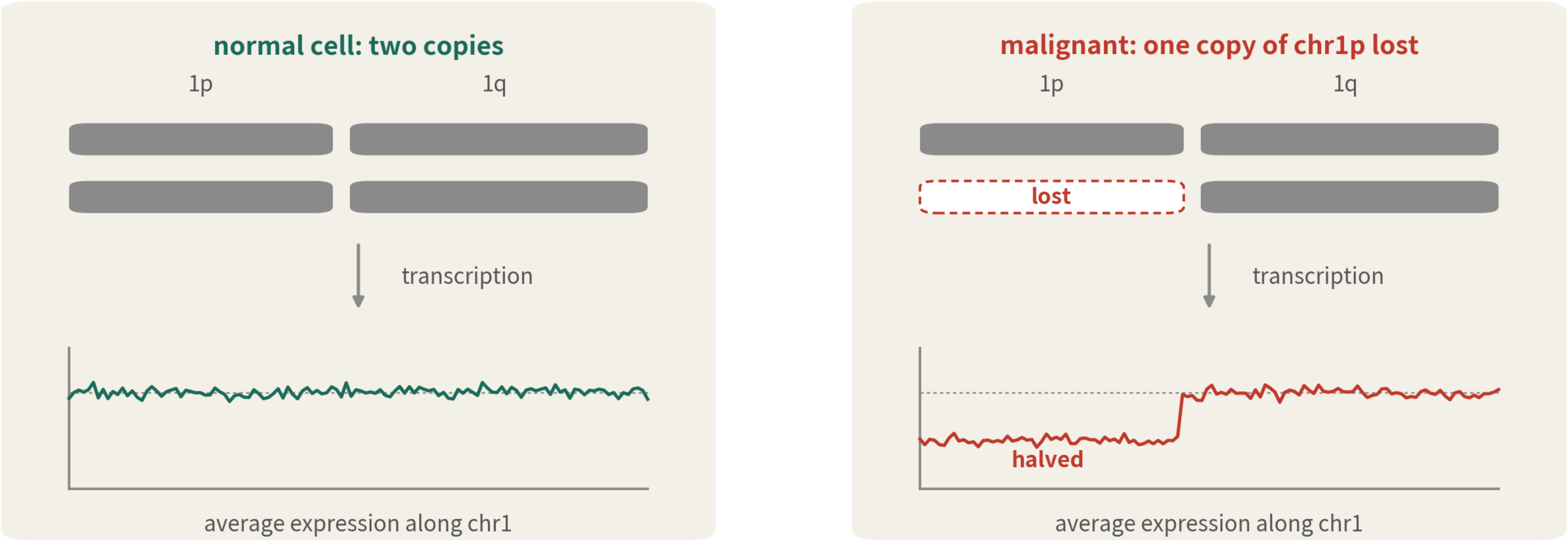
3. Read clone structure off the heatmap, triangulate a verdict, and name three common false signals

01

Where the signal comes from

scRNA never reads DNA — CNV here is inferred, not measured

The defining feature of malignancy is genomic



scRNA never reads DNA — but large gains and losses cast a shadow on segment-average expression

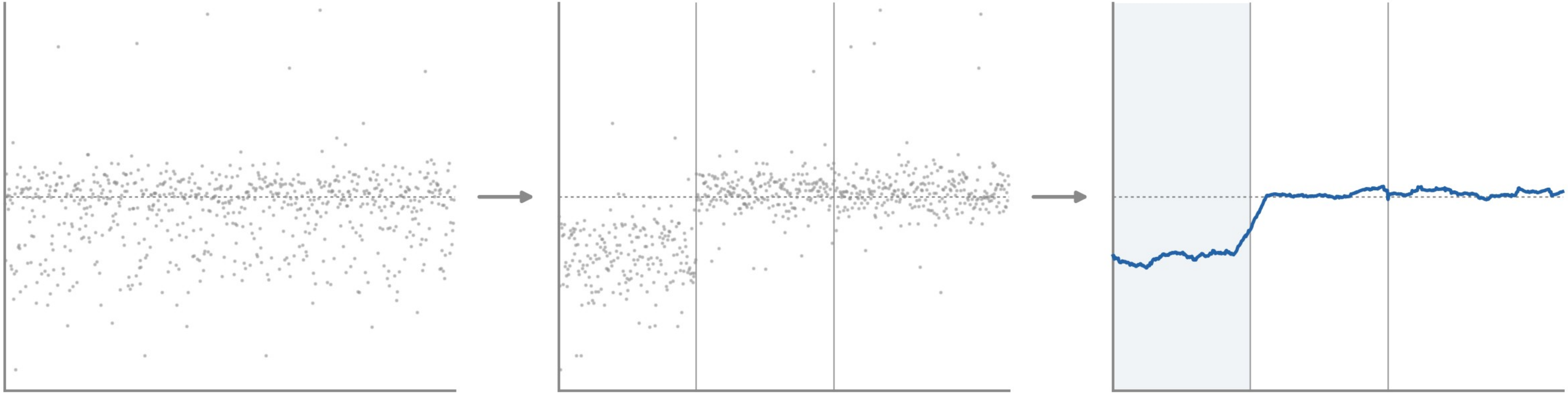
Large gains and losses cast a shadow on segment-average expression

The chain: order, window, baseline

1. genes unordered
per-gene log ratio vs reference

2. sort by genome coordinate
chr1 then chr2, in order

3. sliding-window average (w=51)
average 51 genes per window



pure noise, no visible event

1p dips faintly

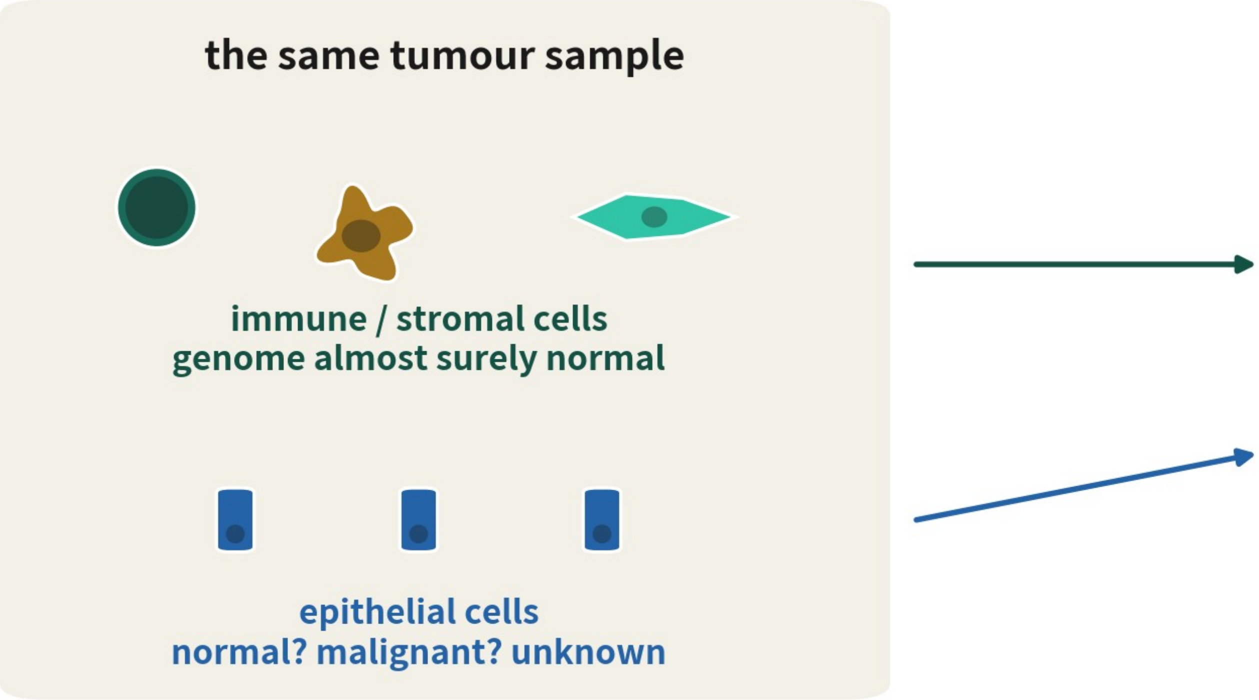
the chr1p loss emerges

simulated data

same numbers: ordering lines the signal up, averaging crushes the noise (~1 log2 unit down to ~1/7)

The same numbers, three steps: genome ordering, windowed averaging, reference-relative shifts

The reference: who counts as normal



reference = the definition of "normal"

observed expression per gene
− reference average
= relative shift (the CNV raw signal)

first choice: immune / stromal cells from the same sample

every colour on the heatmap is relative to this baseline — pick it wrong and the whole map is wrong

First choice: immune / stromal cells from the same sample — their genomes are almost surely normal

CORE IDEA

The CNV signal is a statistical shadow: single genes mean nothing — only segments count.

One gene halving has a hundred explanations; a whole chromosome arm halving usually has one.

02

Three inputs and the example data

counts, annotations, gene order — all three required

inferCNV eats three files

1. counts matrix

	MGH36_A01	MGH53_B02
DLL3	0	12
EGFR	3	0
SOX4	8	21

raw gene-by-cell counts
(extractable from Seurat)

2. annotations

MGH36_A01	malignant_MGH36
MGH53_B02	malignant_MGH53
Mic_93_C07	Microglia/Macro.
Oli_97_D11	Oligodendrocytes

which group each cell belongs to
references are named groups

3. gene order file

WASH7P	chr1	14363
NOC2L	chr1	879584
SAMD11	chr1	860260
...	...	...

chromosome + position per gene
ordering and windows depend on it

```
CreateInfercnvObject(counts, annotations, gene_order, ref_group_names)
```

You already have counts; annotations come from your labels; the gene order file is the new one

Load the package and the built-in example

```r
# BiocManager::install("infercnv") # HMM 功能需要先裝 JAGS
library(infercnv)
set.seed(1234)

counts_f <- system.file("extdata",
  "oligodendroglioma_expression_downsampled.counts.matrix.gz",
  package = "infercnv")
anno_f <- system.file("extdata",
  "oligodendroglioma_annotations_downsampled.txt",
  package = "infercnv")
gene_f <- system.file("extdata",
  "gencode_downsampled.EXAMPLE_ONLY_DONT_REUSE.txt",
  package = "infercnv")
```

system.file points at files shipped with the package — nothing to download

Create the infercnv object

```
anno <- read.table(anno_f, sep = "\t",
          col.names = c("cell", "group"))
head(table(anno$group), 3)

cnv <- CreateInfercnvObject(
  raw_counts_matrix = counts_f,
  annotations_file = anno_f,
  gene_order_file  = gene_f,
  delim        = "\t",
  ref_group_names  = c("Microglia/Macrophage",
             "Oligodendrocytes (non-malignant)"))
```

```
Microglia/Macrophage Oligodendrocytes (non-malignant)
         27                       22
   malignant_MGH36
         33
```

ref_group_names must match the annotation group names exactly — typos error out

What you have at this point

One matrix

Smart-seq oligodendroglioma counts: ~10k genes, ~180 cells

Two identities

two reference groups (microglia, oligodendrocytes) plus tumour observation groups

One map

chromosome and position per gene — ordering and windows depend on it

03

Run it, and the parameters

cutoff, window and denoise each own one job

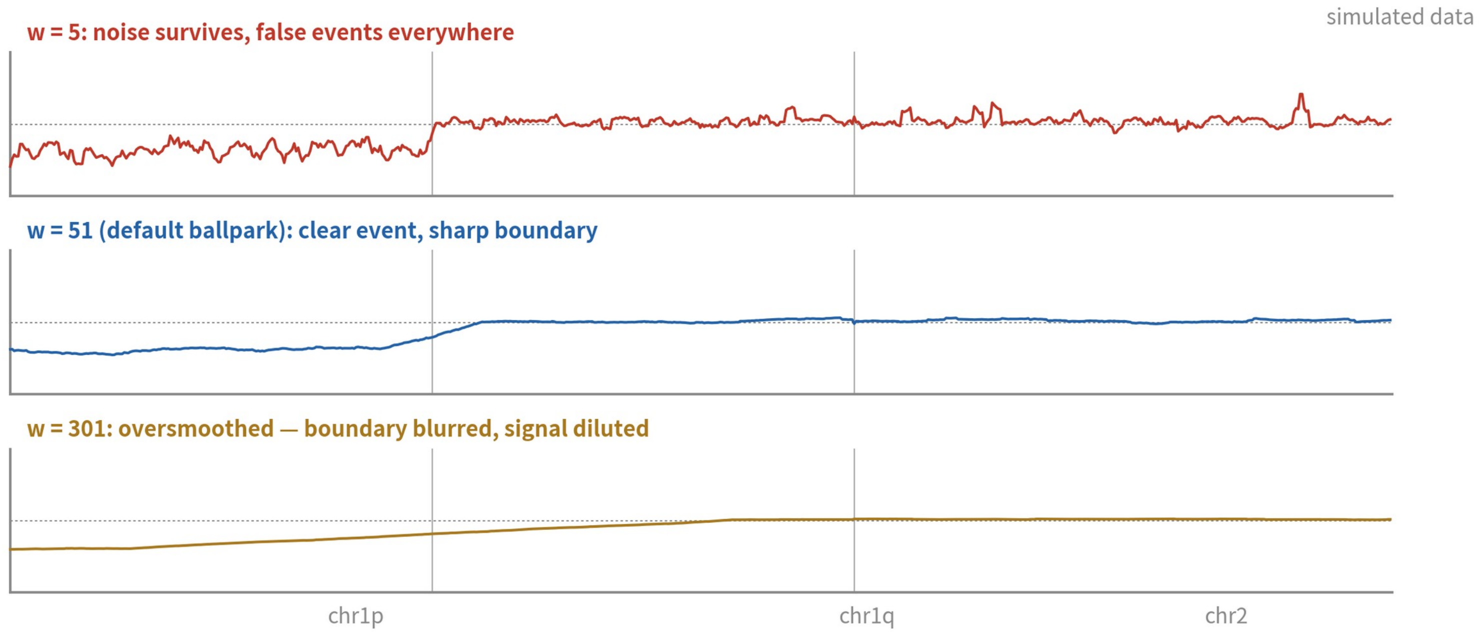
infercnv::run — first pass

```
cnv <- infercnv::run(cnv,
  cutoff = 1,   # 平均計數低於它的基因剔除:
                # Smart-seq 用 1，10x 用 0.1
  out_dir = "output/b16_infercnv",
  cluster_by_groups = TRUE, # 觀察組各自聚類
  denoise = TRUE,       # 壓掉基線附近的白噪音
  HMM     = FALSE)      # 第一輪先看連續訊號
```

```
INFO [...] ::process_data:Start
INFO [...] STEP 15: computing tumor subclusters
INFO [...] Making the final infercnv heatmap
```

It runs for minutes and logs hundreds of lines — the line that matters says the heatmap is done

Three parameters, three jobs



same cells, same genome segment, actually recomputed — only the window width changes

- cutoff drops low-expression genes — noise genes only dilute the window
- window (default 101 genes): too small keeps noise, too large blurs boundaries
- denoise flattens values within the reference's own wobble — a much cleaner map

After the run: inside out_dir

```r
list.files("output/b16_infercnv") |> head(6)
```

```
[1] "infercnv.png"
[2] "infercnv.preliminary.png"
[3] "infercnv.references.txt"
[4] "infercnv.observations.txt"
[5] "infercnv.observation_groupings.txt"
[6] "run.final.infercnv_obj"
```

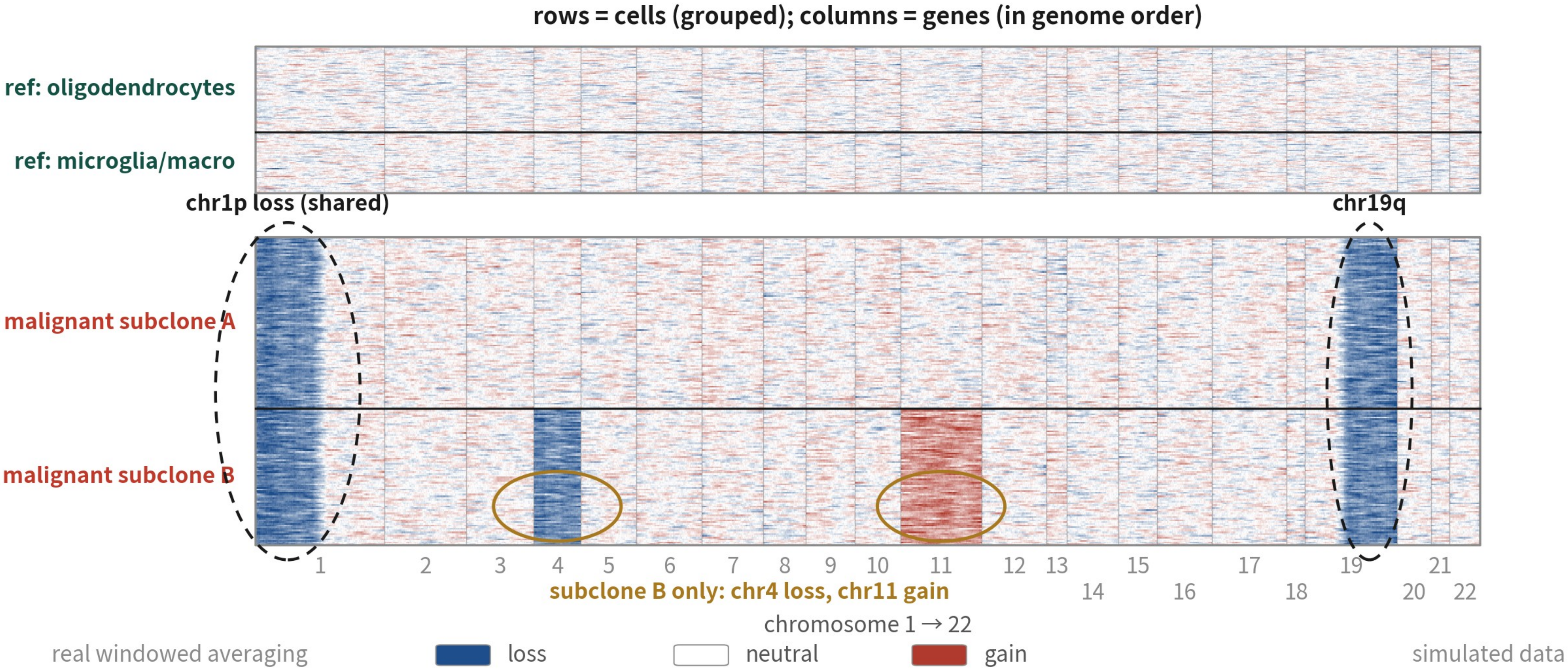
infercnv.png is the main figure; run.final.infercnv_obj is the result object we will read back

04

Reading the heatmap

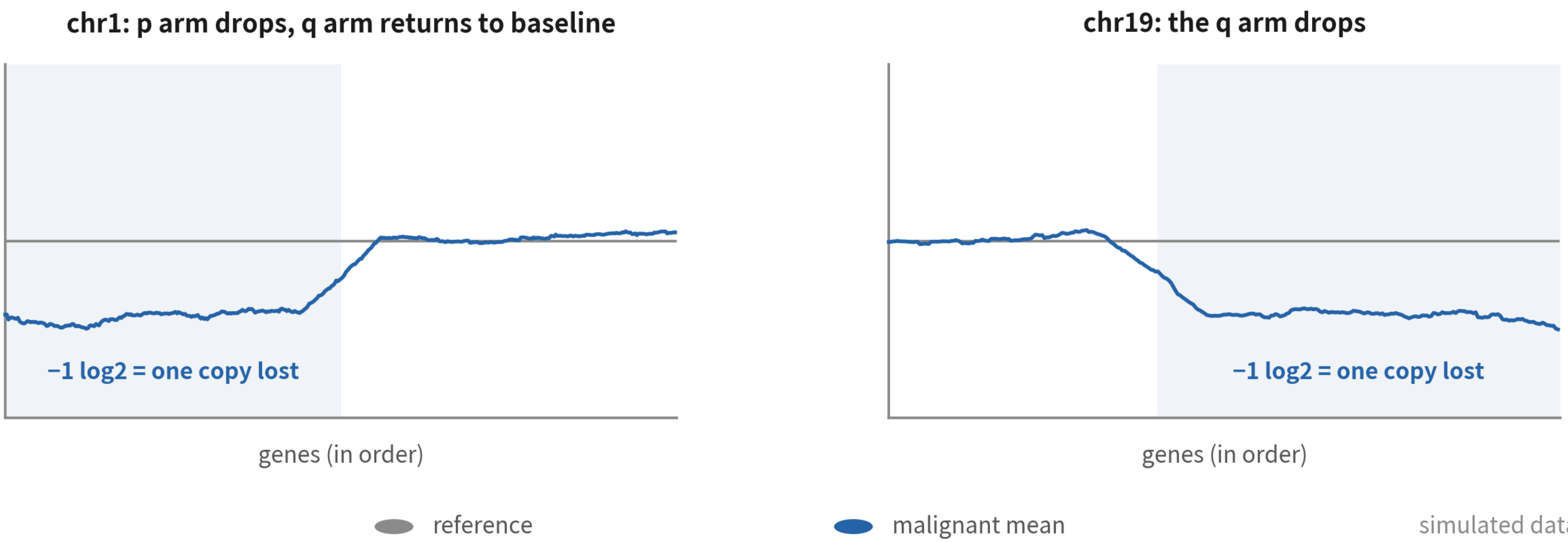
reference first, events second, clones last

Anatomy of the heatmap



The reference half should be near-white; in the observations, only contiguous blue/red blocks are events

The textbook event: chr1p/19q co-deletion

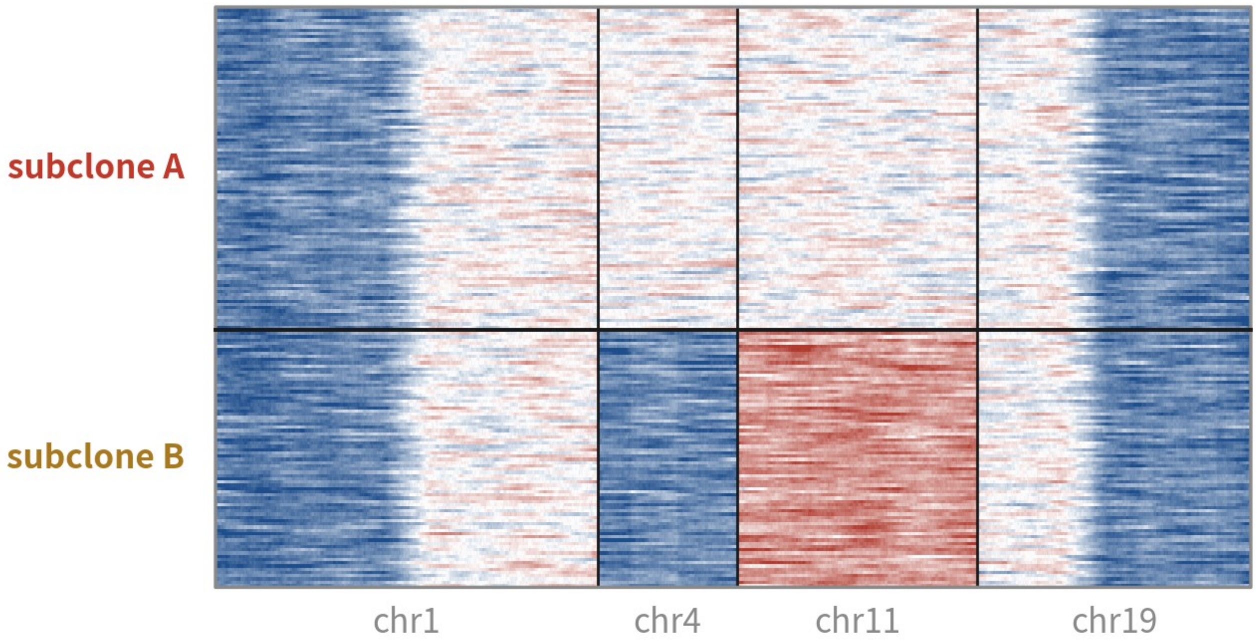


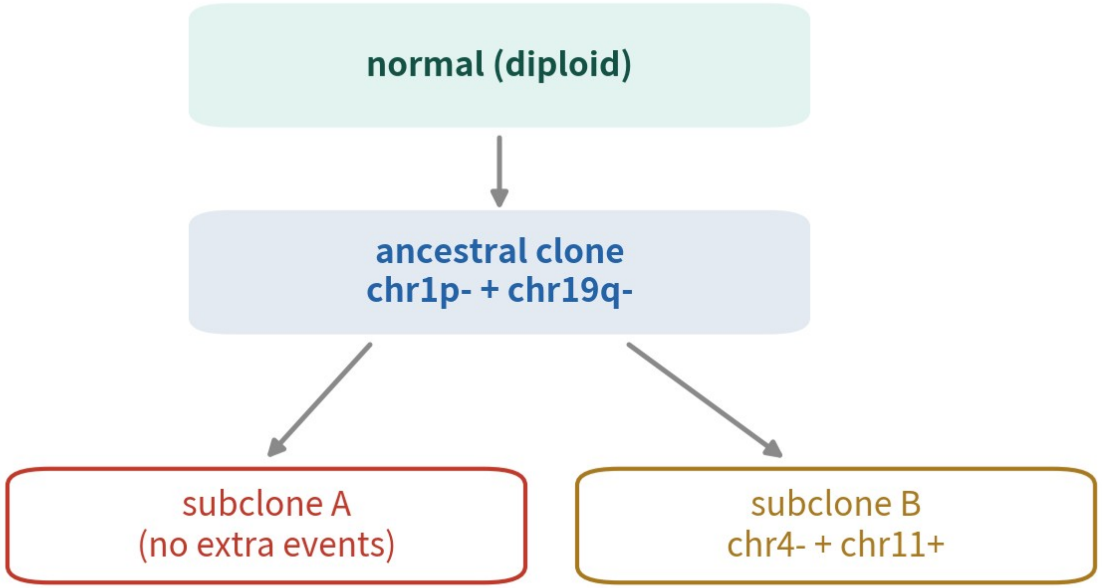
1p/19q co-deletion is the textbook oligodendroglioma event — used clinically for typing

The molecularly defining event of oligodendroglioma — the real example data shows exactly these two

Subclones: the tumour's family tree

four chromosomes (subset of the real result)





shared events form the trunk; partial events form branches

simulated data

Shared events form the trunk, partial events the branches — row clustering lays it out for you

A FIXED READING ORDER

1. Is the reference clean?

2. Any contiguous events? 3. How do they stratify?

If the reference is not clean, nothing downstream counts — the baseline itself is broken.

05

HMM mode and CopyKAT

from shades of colour to discrete calls

HMM mode: colours become states

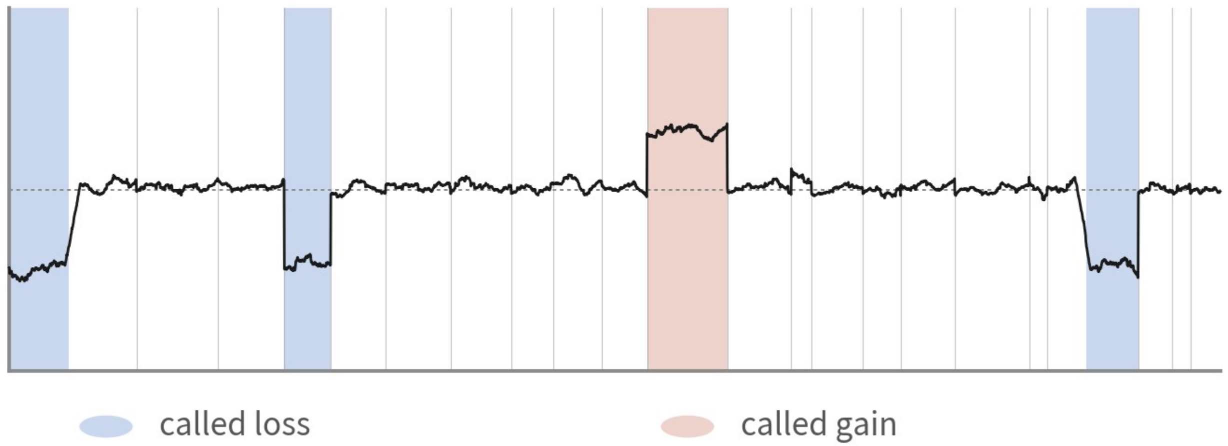
```
cnv_hmm <- infercnv::run(cnv0, # 用未跑過的原始物件
  cutoff = 1,
  out_dir = "output/b16_infercnv_hmm",
  cluster_by_groups = TRUE,
  denoise = TRUE,
  HMM    = TRUE,
  HMM_type = "i6") # i3 = 缺失/中性/增加; i6 再分程度
```

HMM is several times slower — keep it off while exploring, turn it on to conclude

HMM states and per-cell calls

simulated data

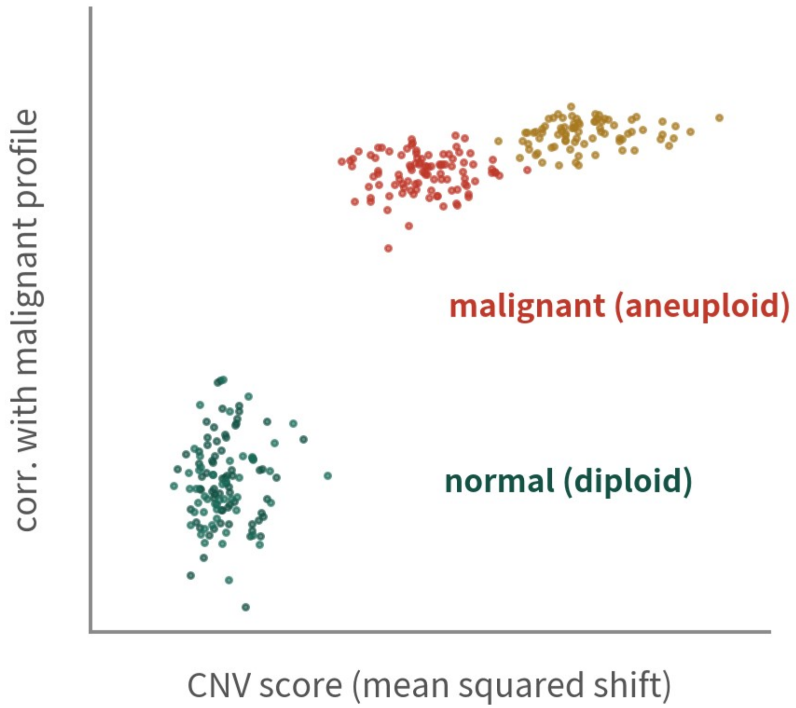
HMM: translate the continuous signal into discrete states (loss / neutral / gain)



subclone B mean signal + a real 3-state Viterbi

inferCNV's i3 mode = these three states; i6 splits gains further,
turning shades of colour into integer copy-number calls

per-cell calls: two clean clouds



CopyKAT's idea: call aneuploid vs diploid per cell, no groups needed up front

Left: continuous signal translated into states. Right: CopyKAT's idea — call aneuploid vs diploid per cell

inferCNV vs CopyKAT

not rivals but complements — people often run both

inferCNV

group-level, great visuals

- you must name the reference groups
- the heatmap communicates; subclone structure is clear
- verdicts live at the group / cluster level

CopyKAT

per-cell, automatic baseline

- no reference needed — it finds diploid cells itself
- outputs aneuploid / diploid per cell
- shines when normal epithelium hides among tumour cells

06

The verdict: triangulate

CNV x clustering x markers, written into metadata

A per-cell CNV score from the result object

```
obj <- readRDS("output/b16_infercnv/run.final.infercnv_obj")
expr <- obj@expr.data          # 基因 × 細胞，中性值在 1 附近
dim(expr)

cnv_score <- colMeans((expr - 1)^2) # 偏移平方的平均
ref_cells <- unlist(obj@reference_grouped_cell_indices)
summary(cnv_score[ref_cells])     # reference 的分數範圍
```

```
[1] 7113 184

    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
0.000876 0.001392 0.001729 0.001846 0.002206 0.004215
```

expr.data is the smoothed, centred matrix — the score is each cell's distance from 1

The triangulation flow

1. CNV evidence
high CNV score,
chromosome-level events + clones

2. clustering evidence
forms its own, often patient-specific cluster
(normal cells mix across patients)

3. marker evidence
lineage markers match
(a carcinoma should look epithelial)

cross-check all three

all three agree → malignant

none present → normal

conflicting → ambiguous: flag it, do not force a call

write the verdict into metadata (malignant / normal / ambiguous) — downstream starts from that column

Any single line of evidence can lie; only the three-way cross-check gets the error rate acceptably low

Make the call, write it into metadata

```
mal_mean <- rowMeans(expr[, -ref_cells]) # 惡性平均 profile
corr_mal <- apply(expr, 2, cor, y = mal_mean)

s_hi <- quantile(cnv_score[ref_cells], 0.95)
lab <- ifelse(cnv_score > s_hi & corr_mal > 0.4, "malignant",
       ifelse(cnv_score <= s_hi & corr_mal < 0.2, "normal",
              "ambiguous"))
table(lab)
```

```
lab
ambiguous malignant    normal
        7       117        60
```

With a Seurat object, hang this lab vector back on with AddMetaData and carry on downstream

What this section did

Quantify

one CNV score and one malignant-profile correlation per cell

Cross-check

score, clustering and markers vote together

Record

malignant / normal / ambiguous lands in metadata — no forced calls

07

Where people get hurt

Three traps — each computed for real

Trap 1: the wrong reference

What it is

Putting a suspect cluster into the reference, or letting tumour cells contaminate it.

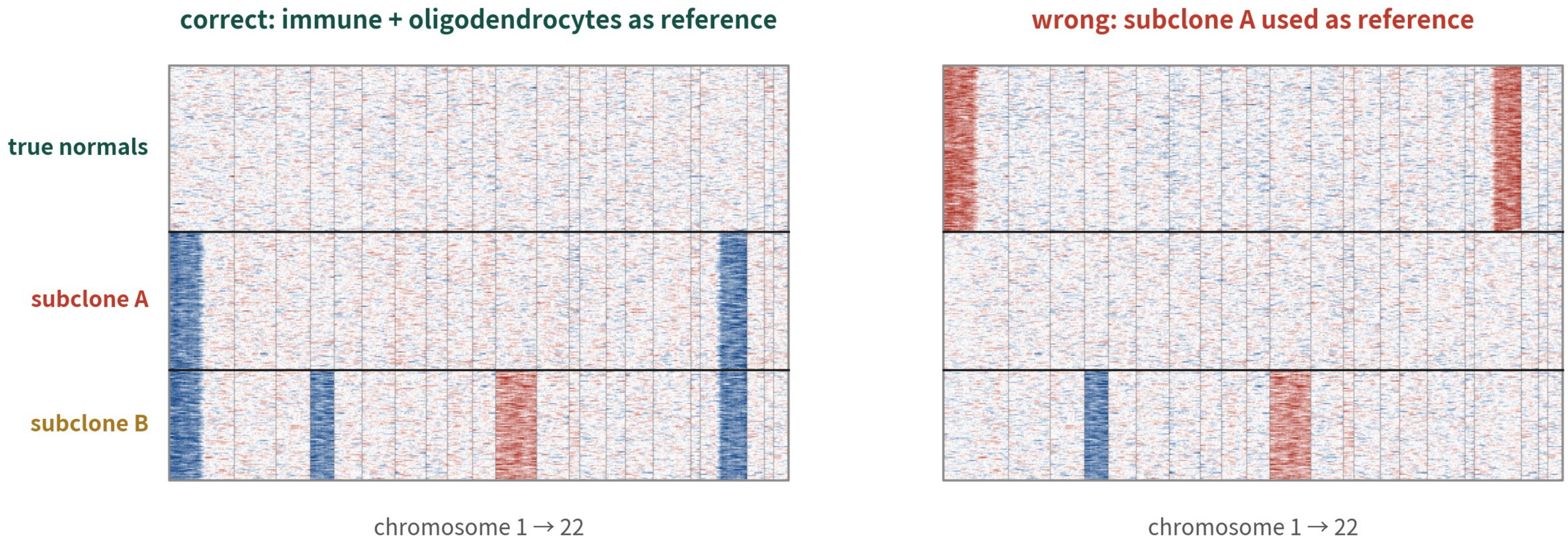
Why it happens

With shaky annotations, the biggest, most normal-looking epithelial cluster gets picked as baseline.

What it costs you

The tumour's CNV hides inside the baseline: the map zeroes out and normal cells sprout fake gains — the signs flip.

Computed: wrong baseline, flattened map



the real events are all visible

malignant rows flatten to zero; normal rows show fake gains — the loss hides in the baseline and the sign flips

simulated data

Same cells, same algorithm, only the reference changed — compare the panels

Trap 2: reading a gene cluster as CNV

What it is

Co-located, co-regulated clusters like HLA or IG loci upregulate together and smooth into a fake amplification.

Why it happens

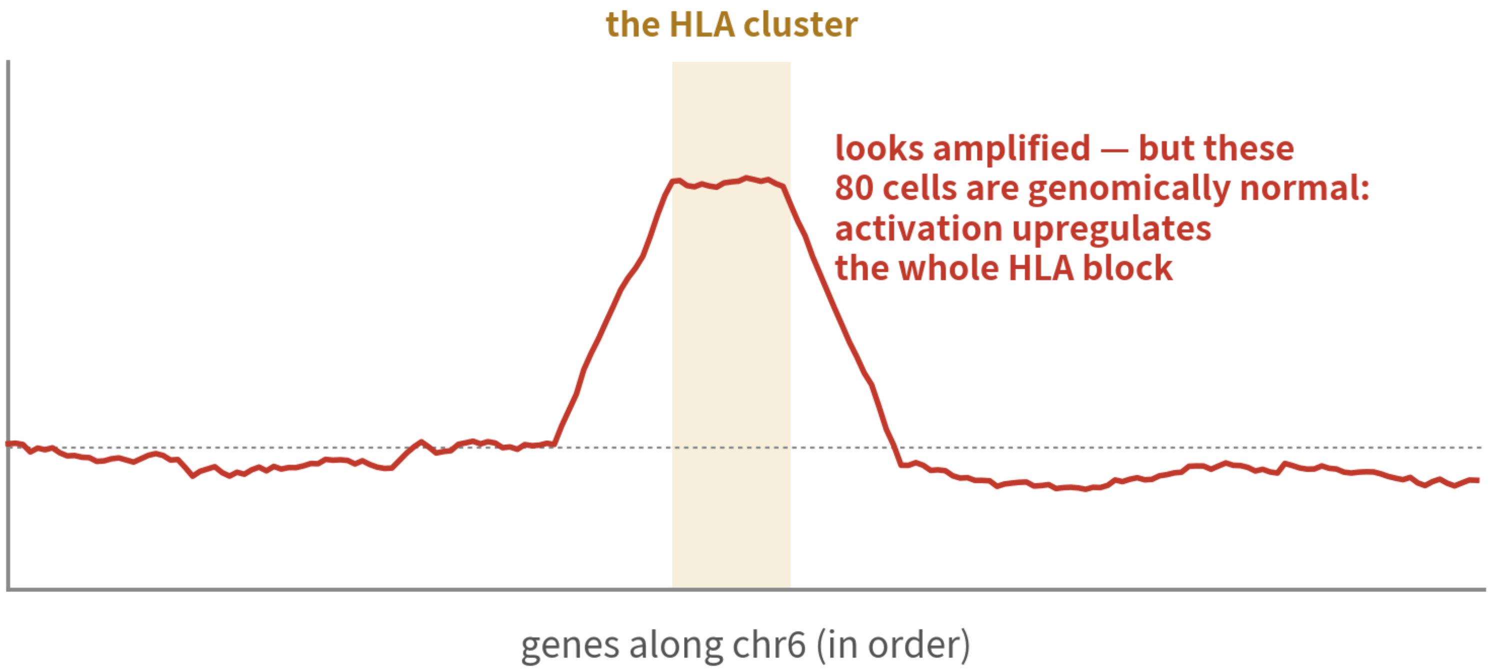
The sliding window assumes neighbouring genes vary independently — co-regulated clusters break exactly that.

What it costs you

Activated immune cells grow a fake chr6p21 event, plasma cells light up an IG locus — and it gets published as a tumour clone.

Computed: the HLA-cluster false signal

simulated data



high-risk gene clusters

HLA (chr6p21)
IG loci (chr2 / 14 / 22)
proliferation / heat-shock blocks

co-located, co-regulated, big shifts
→ smoothing turns them into fake CNV

fix: mask these regions, or never trust them alone

the sliding window assumes neighbouring genes are independent — co-regulated clusters break exactly that

These 80 cells are genomically normal — mask such regions, or never trust them alone

Trap 3: clean CNV, so not cancer

What it is

Seeing a flat CNV profile and labelling the cluster normal, case closed.

Why it happens

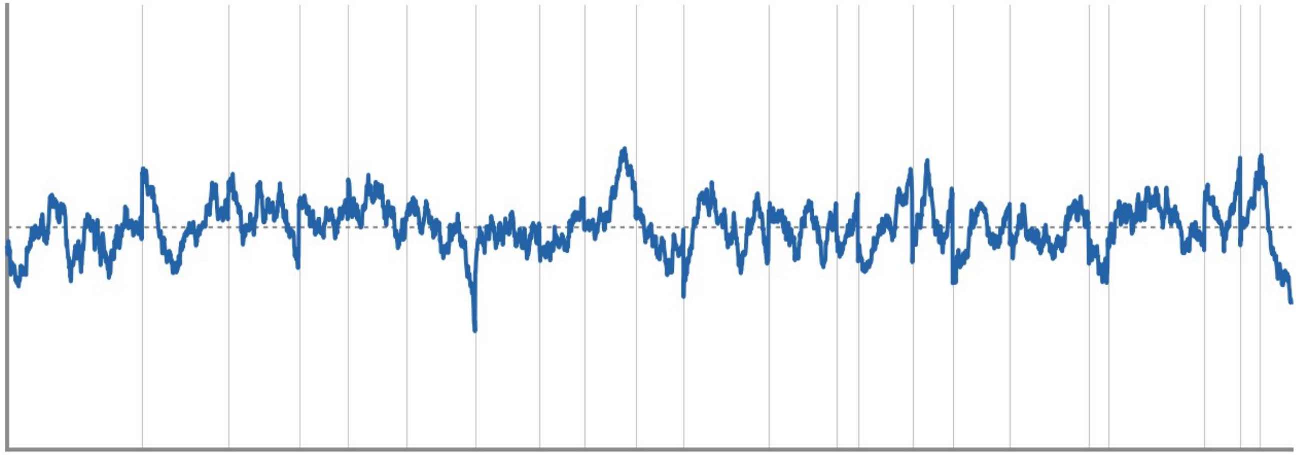
Intuition says cancer means genomic chaos — common, yes; necessary, no.

What it costs you

Near-diploid leukaemias and some paediatric tumours get waved through as normal — downstream is wrong from step one.

CNV is evidence, not the verdict

simulated data



chromosome 1 → 22 (near-diploid cells, computed)

near-diploid cancers are real

- most leukaemias: fusion/point-mutation driven
- some paediatric tumours: strikingly quiet genomes
- near-diploid adult tumours exist too

CNV runs one way: events imply malignant; 'no events implies normal' does not hold

no signal? fall back on mutations, fusions, markers, clinics

CNV is one piece of evidence, not the verdict

The inference runs one way: events imply malignant; no events never proves normal

TRAPS, SUMMED UP

Every colour on that heatmap is an inference relative to the baseline you chose.

State your baseline, your window and your platform — then your CNV conclusions can stand.

5+

Strategy Room: Complex Samples

The standard route is done — this episode you already held a tumour. Now imagine it were a different one

Different tumours, different CNV fingerprints

Look up the fingerprint before you open the map — so you know what to look for, and what 'not found' means

Tumour	Textbook events	Reading decision
GBM	whole-chr7 gain + whole-chr10 loss	whole-chromosome scale — windowing's best case; highly recurrent across patients
Oligodendroglioma	1p/19q co-deletion	arm-level, this episode's example — diagnostic-grade evidence
Breast cancer	focal HER2 (ERBB2, chr17q12) amplification; luminal often 1q+ / 16q-	focal events are narrower than the 101-gene window and get diluted — leave arm-level to inferCNV, corroborate focal amps elsewhere
Liver cancer	1q+ and 8q+ (8q carries MYC)	mostly arm-level gains; the hepatocyte metabolic background is noisy — count only events that recur across patients

Near-diploid tumours: the list and the fallbacks

Whose CNV is often clean

many blood cancers (AML can be near-normal karyotype), some paediatric brain tumours, a few near-diploid solid tumours

Genomic fallbacks

point mutations (full-length Smart-seq reads them; 10x sees only the 3' end), fusion genes, allelic imbalance (Numbat / HoneyBADGER)

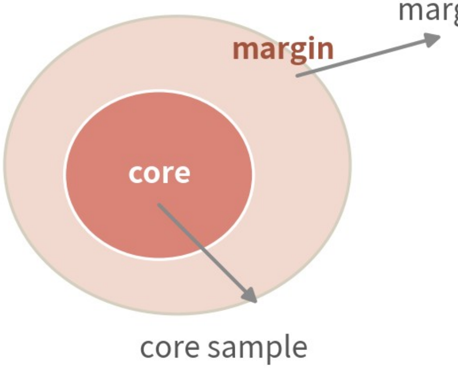
Phenotypic fallbacks

immunophenotype and the differentiation block — compare against a normal marrow / tissue atlas, look for stalled progenitors

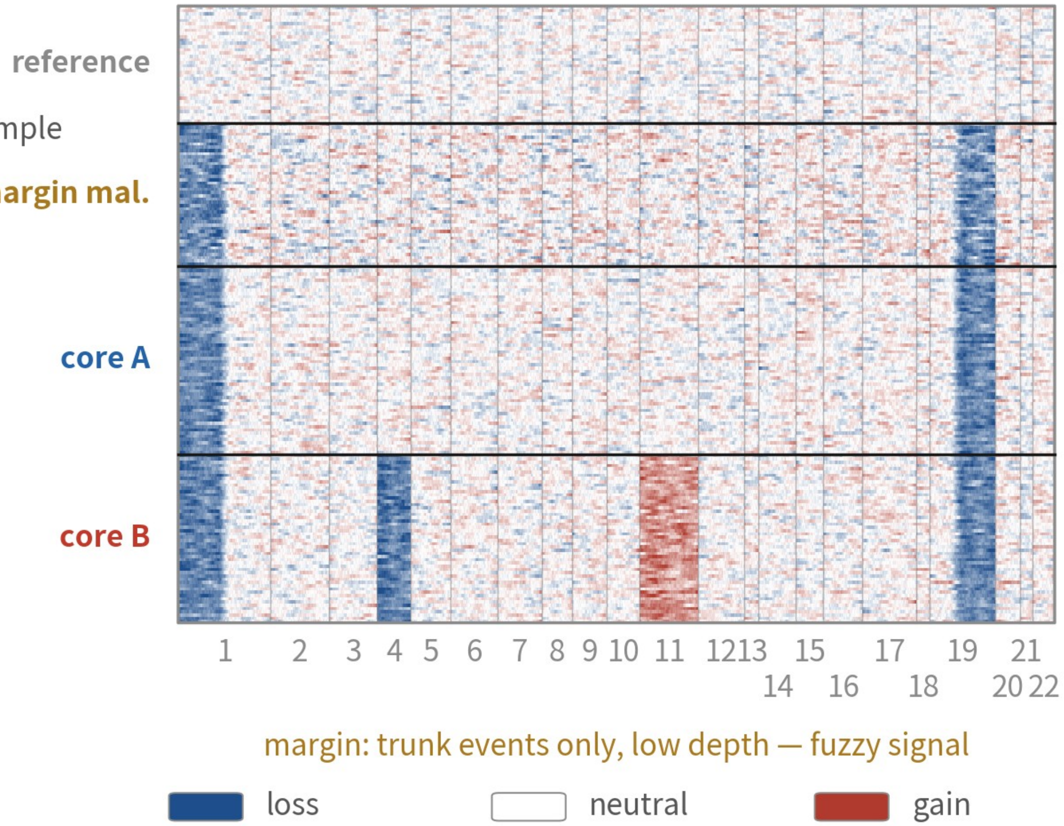
Multi-region sampling: subclones in time and space

simulated data

one patient, two sites



margin + core, same patient (computed)



margin: trunk events only, low depth — fuzzy signal

shared across regions

1p- / 19q-
= early trunk events

private to one region

core B's chr4- / 11+
= late branches

space dates the evolution

margin's extra unresolved
is biology, not failure

Events shared across regions are the early trunk; region-private ones are late branches. The infiltrating margin gives weak signal — extra unresolved there is biology

Triangulation, reinforced: veto, self-check, switch

draft verdict

CNV x clusters x markers (cutoffs from the reference quantiles)

malignant / normal / ambiguous

1. the lineage veto

immune or vascular, yet called malignant?
-> demote to unresolved (doublets or noise)

removes wrong calls only

2. reference self-check

reference cells falling in unresolved ~= your quantile? far above -> contamination

the cutoff audits itself

3. CNV absent? switch

near-diploid tumours: mutations / fusions / allelic imbalance + immunophenotype

no single evidence rules

final labels into metadata

malignant / normal / unresolved — downstream stands on it

the draft is not the end — labels earn their place only after all three checks

(schematic flow; the score and cutoff mechanics are on slides 29-31)

Three insurance checks after the draft verdict: the lineage veto, the reference self-check, and switching evidence tracks when CNV is absent

Episode checklist

- ✓ you can say why clustering + markers cannot call malignancy
- ✓ references come from the same sample's immune / stromal cells
- ✓ cutoff matches the platform: 1 for Smart-seq, 0.1 for 10x
- ✓ read the heatmap reference-first, events second
- ✓ verdicts triangulate CNV, clustering and markers
- ✓ never trust an HLA / IG region 'event' on its own

Tick all six before a malignant label goes into your paper

ONE-LINE SUMMARY

Malignancy calls read the genome's shadow: order, window, baseline — then triangulate.

CNV is strong evidence — provided you know the three places it lies.

Check yourself 1

Why can't clustering plus markers settle malignant vs normal epithelium?

A Clustering algorithms fail on tumour data

B Malignancy is defined genomically; both tools only see the transcriptome

C Epithelial cells have no usable markers

D Malignant cells are too rare

Answer B. Malignant cells share lineage markers with normal epithelium, and clusters reflect expression differences, not genomic state. You need genome-level evidence like CNV inferred from the transcriptome.

Check yourself 2

What is the sliding-window average in inferCNV for?

A Speeding up the computation

B Crushing single-gene noise so chromosome-level coherent shifts emerge

C Correcting batch effects

D Removing low-quality cells

Answer B. Per-gene ratios carry about one log2 unit of noise — no event is visible. Averaging tens of genes per window shrinks the noise several-fold, and contiguous copy-number shifts surface.

Check yourself 3

One cluster: low CNV score, mixes across patients, immune markers. Another: chr1p/19q loss, patient-specific, epithelial markers. Your labels?

A Former normal, latter malignant

B Both ambiguous until DNA sequencing

C Former malignant, latter normal

D The CNV score alone suffices

Answer A. Both are textbook three-way agreements: the first has none of the evidence (normal), the second has all three (malignant). Triangulation means calling only on agreement — and both agree.

NEXT EPISODE

B17: Deconvolution — dissecting bulk with a single-cell reference

By now you can dissect a single-cell dataset in depth. Next we flip it:
300 patients of cheap bulk data and a 3,000-cell reference — let each complete the other.